

safety, and registrants and licensees shall ensure that protection and safety is optimized.” Further details regarding the regulatory body’s responsibilities pertaining to optimization include establishing requirements for optimization, requiring documentation addressing optimization, and the administration of constraints (or thresholds) on dose, or risk, as appropriate. The 2022 IAEA Nuclear Safety and Security Glossary describes the purpose of constraints as boundaries in defining the range of options in optimization.

As discussed previously in this notice, in order to address the problems of subjectivity and over-conservatism that were introduced over time through the implementation of the references to the ALARA principle throughout the NRC’s regulations, the NRC proposes removing these references and replacing them with a requirement for a graded approach to dose management, which would essentially be an application of the concept of optimization that, below the dose limits, relies on various existing regulatory requirements as well as licensees’ individual radiation protection programs to manage dose. NRC guidance would provide an acceptable approach for this. The graded approach to dose management would rely on a series of threshold doses below the regulatory dose limits and dose management actions to be taken at each threshold dose. These threshold doses and dose management actions would generally correspond to existing requirements in 10 CFR part 20 or other regulations and, therefore, should already be incorporated within the radiation protection programs of existing licensees to a degree “commensurate with the scope and extent of licensed activities” as is currently required by § 20.1101.

For any occupational exposure scenario, compliance with 10 CFR part 20 would provide both optimization and reasonable assurance of adequate protection from radiation exposure up to and including planned special exposure events and the proposed planned occupational dose limit extensions. For example, licensees are required to conduct surveys, control access to certain areas, store material appropriately, and use signs, postings, and labels to warn workers of radiological hazards. These provisions are in effect for the full spectrum of radiological hazards that a licensee could encounter during the course of its licensed activities, and thus, these provisions form the first set of dose management actions below the regulatory dose limits that would be

credited as part of a graded approach to dose management.

Under a graded approach to dose management, additional dose management actions would apply as radiological hazards increase and move closer to the applicable regulatory dose limit. The dose levels corresponding to dose management actions are threshold doses in that they represent determinate boundaries above which a specific action is required and below which they are inapplicable (*i.e.*, there is no subjectivity to when a specific dose management action should be taken). Examples of these progressively increasing, threshold doses for occupational exposure are: expected doses of 100 mrem/year, 500 mrem/year, and 5 rem/year. Specifically, above an expected dose of 100 mrem/year, licensees are required to provide instructions to workers (*i.e.*, radiation worker training) per 10 CFR 19.12; above an expected dose of 500 mrem/year, or, more specifically, 10 percent of the applicable limit, licensees are required to monitor doses to individual workers per 10 CFR 20.1502; and above an expected dose of 5 rem/year (*i.e.*, the regulatory dose limit), licensees can exercise the proposed new planned occupational dose limit extension of § 20.1205 or the existing planned special exposure process of § 20.1206, if the situation requires higher dose.

The NRC is developing guidance to further explain and provide acceptable approaches for implementing the graded approach to dose management, including alternative radiation protection measures not already set out in the NRC’s regulations that would help ensure that dose limits are not exceeded. The NRC’s guidance would explain that it would be acceptable for radiation protection measures under the graded approach to dose management to be supported by a comparison of the cost of the radiation protection measure (*e.g.*, shielding, additional workers, robotics) to a reasonably calculated cost-basis of an averted person-rem. The proposed guidance would provide that one example of a reasonable cost-basis standard is provided in NUREG-1530, “Reassessment of NRC’s Dollar Per Person-Rem Conversion Factor Policy.” Specifically, in that guidance document, the NRC establishes the cost of an averted person-rem by multiplying a value of a statistical life coefficient—a factor that corresponds to society’s willingness-to-pay for small reductions in a particular mortality risk—by a cancer mortality risk coefficient. The nominal cost of an averted person-rem under this standard is \$5,200 (in 2014 dollars). Taken together, this means that

it would be acceptable for considering under the graded approach to dose management whether additional radiation protection measures are reasonable based on a need to spend \$5,200 to avoid a person-rem of exposure. Consequently, if a radiation protection measure were more costly than that, the licensee would have an acceptable cost-justified basis for not implementing the measure and instead accruing the dose as long as that dose is within the regulatory dose limits or, if applicable, the licensee complies with the provisions regarding planned occupational dose limit extensions or planned special exposures. In this manner the graded approach to dose management would maintain occupational dose below the regulatory dose limits while replacing the subjectivity of the current ALARA-based regulations with objective cost-benefit analyses. This guidance would be issued subsequent to this rulemaking as part of the NRC’s planned two-phased approach to issuing guidance associated with this rulemaking, see Section VI, “Availability of Guidance,” for more information.

As part of this rulemaking, the NRC also proposes to require a graded approach to dose management with respect to public dose in place of the existing implementation of the ALARA principle. The objective is to maintain a layered protective approach to potential public dose as a precautionary measure. Public dose differs from occupational exposure in several key respects. First, the dose to individual members of the public is generally calculated based on an exposure scenario, whereas occupationally exposed individuals are usually monitored. For example, with respect to nuclear power plant effluents, the member of the public is assumed to be a hypothetical maximally exposed individual who represents the maximum exposure regarding food consumption, occupancy, and other usage in the vicinity of the plant site. Another example is that for a waiting room in a medical facility, the facility may conduct area monitoring and assume conservative occupancy of the waiting room. Another difference between public dose and occupational dose is that the dose limit itself is much lower for the public than for occupationally exposed individuals (*i.e.*, 100 mrem/year vs. 5,000 mrem/year, respectively), and the public dose limit represents a very low level of risk. Specifically, as described in Table 12-4 of the BEIR VII report, the average lifetime risk of dying from cancer is 20 percent, and a lifetime (*i.e.*, 70 years) of

exposure at the public dose limit of 100 mrem/year would conservatively result in an addition of only 0.35 percent to that average lifetime risk. Importantly, the public dose limit is based on the risk of cancer mortality to a large population that is assumed to be exposed at the full limit for a lifetime. However, as just mentioned in the nuclear power plant effluent and medical waiting room examples, in practice, licensees calculate bounding doses to smaller subsets of the population and use the parameters in those calculations to control doses (e.g., reducing effluents from power plants or installing shielding around medical equipment). This means that the actual dose to the average member of the public from NRC licensed activities is assuredly below the limit, and thus, that that individual faces an even smaller risk than the already small risk that is assumed by a lifetime of exposure at the limit.

Because of these inherent conservatisms, the NRC would explain in guidance that one acceptable way of managing dose below the public dose limit—which management would be required under the proposed new graded approach to dose management—is to perform cost-benefit analyses using the assumptions of NUREG-1530, or equivalent assumptions, for any doses to members of the public that are projected to be greater than or equal to 25 percent of the public dose limit (i.e., 25 mrem/year). In addition, this approach to managing public dose would also be acceptable because the existing precautions in 10 CFR part 20 (e.g., waste disposal regulations in subpart K) or other regulations intended to limit public dose (e.g., 10 CFR 50.36a) are sufficient to manage doses to the public within the public dose limits as required by § 20.1101. Stated another way, the assumptions in the calculation of public dose and in the public dose limit itself, in combination with already-existing NRC requirements related to dose management, make it so that it is acceptable to comply with the proposed new graded approach to dose management by not analyzing projected public doses below 25 mrem/year and by performing a cost-benefit analysis for projected public doses greater than or equal to 25 mrem/year. This approach would essentially reestablish the original intent of the ALARA principle of minimizing dose below limits to the extent that doing so is reasonably achievable and would ensure that dose limits are not exceeded. Again, as with the implementation of the proposed new graded approach to dose management for occupational dose,

whereas the NRC's proposed guidance provides one way by which a licensee can satisfy that requirement for public dose, licensees may propose other ways to satisfy the requirement.

Effluents

The NRC's regulations include requirements for maintaining control over the release of radioactive material to the environment during normal reactor operations. For example, under § 50.34a and § 50.36a, the NRC requires nuclear power plant licensees to include in their facilities measures to control radiological effluents to the environment—including via monitoring and control systems—and to have in their licenses technical specifications to control the release of effluents. For power reactors, Appendix I to 10 CFR part 50 provides numerical design objectives regarding effluents. These design objectives are translated into performance criteria that are reflected in plant-specific technical specifications. In these requirements, effluents are quantified using the calculated dose that a member of the public would receive when exposed to the effluents under limiting conditions, i.e., the hypothetical maximally exposed individual who represents the maximum exposure regarding food consumption, occupancy, and other usage in the vicinity of the plant site.

A similar requirement regarding air emissions for licensees not subject to § 50.34a and § 50.36a is provided in § 20.1101(d). These requirements were added to the NRC's regulations to provide design objectives and constraints to ensure that radioactive effluents (and thus the resulting public doses) would be maintained consistent with the ALARA principle. Additionally, the NRC has used these types of regulations to meet environmental protection-related obligations under the Clean Air Act (CAA) and to demonstrate compliance with the environmental protection standards for nuclear power operations under 40 CFR part 190. In NUREG-0543, the NRC describes how a licensee would be in compliance with the dose-based requirement in 40 CFR part 190, if the licensee maintains effluents below the numerical criteria of Appendix I to 10 CFR part 50.

In the context of the CAA, the NRC has historically worked with the U.S. Environmental Protection Agency (EPA) to develop effluent standards that are sufficiently low to support EPA determinations and to ensure that NRC licensees are not subjected to redundant regulation from multiple agencies (see, e.g., 42 FR 2858, 54 FR 51654, and 61

FR 65120). As part of the development of the current air emissions constraint in § 20.1101(d), the NRC ensured that the value of that constraint would be such that the Administrator of the EPA could determine that the constraint provided “ample margin of safety,” as is required under Section 112(d)(9) of the CAA. This “ample margin” determination is explained in the proposed and final rules that promulgated the National Emissions Standards for Hazardous Pollutants (NESHAPs) for radionuclides (54 FR 9612 and 54 FR 51654, respectively). EPA supported its determination that the NRC's regulations would satisfy the “ample margin” statutory requirement with studies of air emissions from NRC and Agreement State licensees. In total, these studies considered air emissions from 412 facilities on an annualized basis. EPA found that air emissions from most facilities do not result in doses exceeding 1 mrem/year with a small percentage of facilities approaching, but none exceeding, 10 mrem/year.

As part of its response to E.O. 14300, the NRC reconsidered risk analyses that are based on the LNT model and the implementation of the ALARA principle, as described elsewhere in this proposed rule. As it pertains to doses to members of the public, the NRC maintains that there is reasonable assurance that public health and safety is adequately protected at all doses below the NRC's current regulatory dose limit of 100 mrem/year. Dose reduction below this limit in and of itself is not necessary for ensuring the public health and safety; instead, as clarified by this proposed rulemaking, dose reduction serves the purpose of ensuring that ample margin exists to the regulatory dose limit and thus ensures that the limit is not exceeded. Regarding the contribution of effluents to public dose, this margin to the dose limit is maintained by the requirements in the NRC's regulations that pertain to the monitoring and control of effluents and by licensees' actions to manage dose, which could include performing cost-benefit analyses to support decision making on additional measures for controlling doses below the dose limits.

Accordingly, the NRC is proposing revisions to radionuclide emissions standards in 10 CFR 50.34a, 10 CFR 50.36a, 10 CFR part 50 Appendix I, and 10 CFR 20.1101(d) because it has determined that they are excessively cautious and overly burdensome. Specifically, the NRC is proposing to increase its radionuclide emissions standards from the current regulatory constraint in 10 CFR 20.1101(d) of 10 mrem per year to 25 mrem per year. The

NRC's position is that this would remove excess conservatisms while continuing to provide an adequate basis to EPA that the NRC's regulatory framework provides "ample margin of safety to protect the public health" under section 112(d)(9) of the CAA. To illustrate, in its proposed NESHAP rule for radionuclides, the EPA characterized the maximum lifetime risk of fatal cancer from natural background radiation from all sources, including naturally occurring radon, as approximately 1×10^{-2} , or 1 case per 100 people. Using a current, widely accepted, and likely conservative cancer risk coefficient of 5×10^{-4} per rem (see NCRP 180, Section 4.1), an individual receiving a dose of 10 mrem per year, the current regulatory constraint in 10 CFR 20.1101(d), for 70 years would experience an excess fatal cancer risk of 3.5×10^{-4} , or about 0.04 cases per 100 people. If the dose to that individual were raised to 25 mrem per year, the proposed new regulatory constraint, for 70 years, the individual would experience an excess fatal cancer risk of 8.7×10^{-4} , or about 0.09 cases per 100 people. When compared to the baseline lifetime risk of fatal cancer of approximately 0.2 (*e.g.*, as provided in Table 12-4 of the BEIR VII report), or 20 cases per 100 people, the risks of these environmental levels of radiation exposure, at both 10 mrem per year and 25 mrem per year, are a small fraction and well below the 100 mrem per year public dose limit.

In addition to reconsidering its current radionuclide emissions standards through a risk perspective, the uncertainties associated with risk estimates based on extrapolations from high-dose and high-dose-rate data provide additional support for the NRC's position that its proposed increase to its radionuclide emissions standards would remove excess conservatisms while still providing reasonable assurance of adequate protection of the public health and safety and would also continue to support EPA's determination that the NRC's standards provide an ample margin of safety under the CAA. Adjustments in the slope of the dose response curve, such as is done with the DDREF, are helpful in the extrapolation of high-dose/high-dose-rate data to low doses; however, there is subjectivity and potential conservatism associated with this adjustment. Additionally, there is evidence for adaptive cellular response, which would mitigate the health effects of exposures at low doses, especially those resulting from effluents. The NRC determined that these uncertainties

were undervalued in the establishment of the radionuclide emissions standards that the NRC currently uses.

The NRC's proposed approach to the regulation of effluents would involve allowing licensees to continue using the existing effluent constraints of 10 mrem per year or allowing licensees to use a new constraint of 25 mrem per year TEDE or TED, as applicable. Regardless of the constraint used by a licensee, if the licensee demonstrates that its effluents are below the 25 mrem per year level, it would only be required to collect and retain effluent data on an annual basis and in a format that can be inspected by the NRC. If a licensee releases effluents greater than or equal to the 25 mrem per year constraint, that licensee would be required to collect and retain effluent data and submit relevant reports to the NRC on an annual basis until levels are returned to below the 25 mrem per year constraint. Additionally, such a licensee would be required to evaluate and consider implementing cost-justified corrective actions to restore effluent levels to below the 25 mrem per year constraint. NRC guidance would provide that one acceptable method for performing this evaluation would be to use the dollar per person rem value from NUREG-1530. If a cost analysis demonstrates that corrective actions are not justified, the licensee could propose a new constraint that would support continued operations in a cost-justified manner, but this new constraint would be required to be below the public dose limit. A similar approach would be taken during the licensing of a new facility, *i.e.*, a constraint higher than the regulatory constraint, but lower than the public dose limit, could be proposed as needed to support operations in a cost-justified manner. These changes are being proposed to §§ 20.1101(d), 50.34a, and 50.36a and to appendix I to 10 CFR part 50. To avoid disruptions to existing licensees, the NRC is proposing these changes such that existing effluent programs will remain compliant with the NRC's requirements, as amended, and that licensees can adopt changes on a voluntary basis.

The NRC understands that its radionuclide emissions constraint relates to EPA's ample margin determination under section 112(d)(9) of the CAA. According to CAA section 112(d)(9), the EPA must consult with NRC prior to a new or revised ample margin determination. The NRC's proposed position is that the proposed change to its radionuclide emissions constraint could continue to provide an adequate basis to EPA for such a determination.

Additionally, the NRC is proposing to revise paragraph C of Section IV of Appendix I to 10 CFR part 50 to add "§ 52.110" to the applicability of the provisions in that paragraph. This is an editorial change to make paragraph C consistent with the applicability specified in the introductory paragraph of Section IV.

Planned Occupational Dose Limit Extension

In the NRC's current regulations, planned special exposures (PSEs) (see 10 CFR 20.1206) allow occupationally exposed individuals to receive doses in excess of the applicable limit to a maximum of twice the applicable limit in one year, provided that certain criteria are met (*e.g.*, documentation tracking lifetime dose and remaining bank of PSE-dose). There is a cap on PSE-dose of five times the applicable limit over the lifetime of an individual. To ensure compliance with this lifetime cap, a licensee must ascertain the lifetime exposure history of an individual prior to conducting a PSE of that individual. Additionally, a PSE is viewed as a tool to be used only during exceptional circumstances. As such, PSEs involve additional reporting and recordkeeping requirements when compared to routine occupational exposures. Since the NRC added the regulations allowing for PSEs in the 1991 revisions to 10 CFR part 20, PSEs have not been used by licensees, as demonstrated through a lack of reports having been submitted to the NRC per § 20.2204.

The NRC has determined that the administrative burden associated with PSEs (*e.g.*, determination of lifetime exposure histories and additional reporting requirements) combined with the characterization of a PSE as a tool to be used only in exceptional circumstances is not commensurate with the radiological risk involved with exposures at occupational levels. Additionally, the increased administrative burden associated with PSEs likely dissuades licensees from viewing PSEs as a viable option for occupational dose management. Given this background and to enable flexibility in the balancing of occupational exposure with operational needs, the NRC is proposing to codify in 10 CFR 20.1205 a new process for allowing workers to receive doses in excess of applicable annual limits: the planned occupational dose limit extension (DLE). This optional process would make available to licensees a method to manage a justified, pre-planned exceedance of annual occupational limits for workers, provided that an

adequate decision-making process is applied to support its use and that the overall dose is limited within specified multi-year average values and annual limits are limited to twice the applicable limit. Although the NRC is also maintaining the current PSE process in its regulations, the proposed new planned occupational DLE would be less burdensome for licensees to implement while still maintaining occupational doses, and thus the underlying risk, within acceptable values.

The proposed new planned occupational DLE would allow licensees to access, in the current year, occupational dose that was not given to a worker in previous years. Dose limits generally serve two purposes: (1) to avoid nonstochastic/deterministic effects, also called tissue effects, and (2) to manage stochastic risk to an acceptable level. Nonstochastic/deterministic effects, or tissue reactions, are effects that are only seen once a threshold is exceeded and their severity is dependent upon the dose received. Protraction of dose reduces the risk of nonstochastic/deterministic effects especially at doses below the threshold because the body is able to heal the biological damage resulting from the dose received. Stochastic effects are random in nature; they are not subject to a threshold and the severity of the health effect is independent of the dose received. Currently accepted models assume that for stochastic effects the likelihood/risk of an adverse health effect occurring increases proportionately with dose. Each of the NRC's occupational dose limits functions to limit the risk of adverse health effects associated with radiation exposure to levels that have been determined to be acceptable for routine occupational situations.

Occupationally exposed individuals rarely approach even small fractions of applicable dose limits (see NUREG-0713) because of licensees' existing dose management efforts, which are generally founded on ALARA practices. In general, the fact that the risk from radiation exposure has been made negligible because of these dose management efforts is a net benefit. However, it is likely that these measures have resulted from overly conservative decision-making. Additionally, some licensees use locally developed administrative limits to maintain margin to occupational dose limits. Occasionally, normally during maintenance periods, licensees may encounter the need for workers to be exposed to doses in excess of annual limits but still within the standards for

protecting against the health effects of radiation exposure (e.g., long-term averages with respect to lifetime doses). In these cases, in part to avoid the added burden of PSE use, licensees employ additional measures (e.g., shielding) or use additional workers to spread out the dose so as to maintain individual doses below limits. However, these actions may increase the cost and the time associated with the work in a manner that is not commensurate with the risk associated with the dose.

To encourage more reasonable decision-making regarding doses at levels corresponding to occupational exposures, consistent with the original intent of the ALARA principle and the proposed graded approach to dose management, the NRC is proposing to add to its regulations the planned occupational DLE process to allow licensees to periodically exceed applicable dose limits for an individual worker provided that certain criteria are met. The NRC proposes to define the dose allowed for a planned occupational dose limit extension as the occupational exposure that was unused by the individual over the most recent 5-year period. For example, for the TEDE limit of § 20.1201(a)(1), this would mean the dose available is equal to the product of 5 years and 5 rem TEDE, totaling 25 rem TEDE, minus the actual annual TEDE received by the individual in the current year and the preceding 4 years. A similar approach can be used to determine the allowable dose for a planned occupation dose limit extension of the deterministic limits, with the exception of the lens dose limit. Licensees would be able to apply this allowable dose, or "retrospective dose," up to a total dose of twice the applicable annual dose limit in the current year. Thus, the annual limit for the planned occupational dose limit extension would be consistent with the dose that is allowed through the planned special exposure, with the exception of lens dose.

The safety basis for the proposed new planned occupational dose limit extension process relies on the fact that annual dose limits are derived with the intent to limit total lifetime exposure to an individual worker and to preclude deterministic effects. However, because total lifetime exposure is not a practical value to measure, radiation protection standards prescribe annual stochastic limits that are essentially fractionated lifetime totals. Therefore, the annual stochastic limits in and of themselves should not be viewed as demarcations of safety or thresholds above which health effects are expected. Instead, they are regulatory tools to manage the long-

term risks of exposure. As such, these limits can safely be exceeded to a certain extent in the short-term, provided that long-term doses continue to be controlled adequately. When an occupationally exposed individual receives an annual dose below the annual limit, that individual is experiencing a smaller risk than was originally assumed to be acceptable for a radiation worker in developing the annual dose limits. In general, this is a positive outcome, primarily because of the corresponding reduction in risk that is associated with reductions in dose inherent to the statistical nature of stochastic effects. However, this also means that there often exists unused retrospective dose that could be safely used, provided that an adequate decision-making process is applied to support its use and that the overall dose is limited within specified multi-year average values. Deterministic effects would not result from the planned occupational dose limit extension process because those effects are only seen when certain thresholds are exceeded, and the restrictions on the planned occupational dose limit extension process would maintain doses below those thresholds.

The proposed new planned occupational dose limit extension process, as described in § 20.1205, would allow a licensee to authorize an individual worker to receive a dose in excess of annual occupational dose limits provided that (1) the licensee does not authorize a dose that would cause the individual to receive twice the applicable annual dose limit in a year, and (2) sufficient retrospective dose is available to the individual. This process would also entail limitations and reporting and recordkeeping requirements. For example, approval of a planned occupational dose limit extension for an adult worker must be in writing before the exposure occurs; the individual must be informed of the purpose of the planned operation, estimated doses and their associated risks, and measures taken to manage doses; and the licensee must determine the occupational exposure of the individual during the current and preceding four years (see 10 CFR 20.1205). Furthermore, declared pregnant women and minors would not be allowed to participate in planned occupational dose limit extensions. Additionally, because of the uncertainty associated with the health risks of lens dose, as discussed previously in this document, the NRC has decided to exclude exceeding the annual lens dose limit from the proposed new planned

occupational dose limit extension process; however, licensees could still use the current PSE process, which will remain in the NRC's regulations (including the required lifetime dose determination), in situations that involve a need to exceed the annual lens dose limit. Finally, the requirement to implement a graded approach to dose management would also apply to the management of the doses received during planned occupational dose limit extensions such that unnecessary occupational exposure would be avoided and radiation protection practices would be employed commensurate with the risks involved.

Public Dose Limits

The NRC is not proposing to change its current public dose limit, which is 100 mrem per year. The NRC considered several recommendations to change the public dose limit. As described previously, the 2015 PRMs requested an increase in the public dose limit based, in part, on the assertion that radiation exposure is beneficial. Additionally, the Idaho National Laboratory—a U.S. National Laboratory dedicated to energy research and development—recently suggested in a public report (“Reevaluation of Radiation Protection Standards for Workers and the Public Based on Current Scientific Evidence,” INL/RPT–25–85463, Revision 0, July 2025) that the public dose limit be raised, in part, to increase public acceptance of radiation exposure.

The NRC regulates the civilian use of radioactive materials in a manner that provides reasonable assurance of adequate protection of the public health and safety. Although the current public dose limit is sufficient to provide reasonable assurance of adequate protection of the public health and safety, the NRC understands that it is a generic limit and, thus, may be overly conservative in certain, specific circumstances. The NRC also recognizes that a number of factors may weigh in favor of allowing a higher public dose limit on a case-by-case basis when such conservatisms are specifically identified and accounted for, including why such an allowance would remain protective of the public health and safety. As discussed previously, in response to E.O. 14300 section 5(b), the NRC reexamined its approach to radiation protection and recognizes that there are inherent limitations in the LNT model, particularly at low doses, that could be better addressed in the NRC's rules. One example is the decision in the NRC's 1991 rulemaking to disallow exceptions to the public dose limit for new applicants. Upon reexamination, such a

limitation is not needed for reasonable assurance of adequate protection of public health and safety and is also inconsistent with the principles of the E.O.s discussed in this document concerning enabling the use of nuclear power, particularly in light of the declared energy emergency in E.O. 14156. Taken together, these factors weigh in favor of the NRC reconsidering whether exceptions to the generic public dose limit of 100 mrem per year may be allowed. Therefore, although it is not proposing to change its current public dose limit or define specific exceptions to that limit, the NRC is proposing revisions to the regulations in § 20.1301 that would allow a licensee or an applicant to request alternative public dose limits on a sufficiently supported, case-by-case basis.

First, in § 20.1301(b), the NRC is proposing a path to enable a licensee or applicant to request prior NRC authorization for a public dose limit in excess of 100 mrem per year for members of the public who have access to controlled areas. Such requests may be on a generic basis (*e.g.*, for a design). A controlled area, as defined in § 20.1003, is an area, outside of a restricted area but inside the site boundary, to which access can be limited by the licensee for any reason. The NRC's current regulations extend the public dose limit to all areas within the site boundary. However, this approach may be excessively restrictive if dose is appropriately managed in the controlled areas. This is because members of the public do not maintain a lifetime of continuous occupancy in spaces within the controlled area boundary of licensed facilities. Instead, members of the public typically access these spaces on a temporary basis for such non-occupational-exposure purposes as tours, awaiting for or accompanying those receiving medical treatments, or work activities separate from those of the facility like making deliveries to/from the facility. Further, it is reasonable to assume that the majority of these individuals will be adults and that the time of exposure of any children will be small when compared to the lifetime of exposure that is considered when determining acceptable risks for stochastic health effects.

Therefore, the NRC is proposing to amend its regulations to provide that, as long as a licensee or applicant establishes appropriate dose management measures (*e.g.*, signage, information briefings and area monitoring to ensure compliance with the proposed higher annual dose limit), a higher public dose limit within the

controlled area may be approved on a case-by-case basis. The NRC's case-by-case review would consider such things as the likely cumulative exposure of a member of the public under the proposed new dose limit given the time that the member of the public is expected to be within the controlled area and the effectiveness of the proposed dose management measures.

Second, in § 20.1301(d), the NRC is proposing to remove references to ALARA, to remove the upper limit on the annual dose limit for a member of the public that may be requested (which is currently set at 500 mrem per year), and to specify the information that is required in an application by a licensee or applicant for prior NRC authorization, on a case-by-case basis, for a public dose limit in excess of 100 mrem per year. The regulation at § 20.1301(d) was originally intended to provide a process for facilities existing at the time of the regulatory changes implemented in 1991 that found it difficult to meet the then newly enacted public dose limit of 100 mrem per year (see 56 FR 23360). However, the NRC now proposes to clarify that any licensee or applicant can apply for a higher public dose limit by providing the information required by the regulation, which the NRC will review on a case-by-case basis. Specifically, such an application must: (1) demonstrate the need for and the expected duration of operations in excess of the public dose limit; (2) describe the licensee's program to assess and control dose within the proposed higher limit; and (3) provide a supporting basis for the proposed higher limit, including why it remains protective of the public health and safety. Such a request for a higher public dose limit for the unrestricted area would require a more detailed analysis than a request for a higher public dose limit for the controlled area because of the lack of control that a licensee can exert over the unrestricted area and the presumably larger population that could potentially be subject to the proposed higher doses. Because of the statistical nature of stochastic health effects, the larger the population that receives a given dose, the larger the potential health impact. Therefore, it is likely that the NRC would generally reserve approvals for higher public dose limits that are limited in duration and/or demonstrated to have limited population impacts.

In their applications, in order to demonstrate that the requested higher public dose limit remains protective of the public health and safety, licensees and applicants could apply the critical

group or representative person concepts described in ICRP Publication 101, and earlier ICRP references, to perform prospective dose assessments.

Determinate or probabilistic assessments, or a combination of both, could be used to support the request. Habit data and physiological characteristics should be representative of the affected population and not overly conservative in terms of assumptions. Additionally, licensees and applicants may consider the use of dose constraints, additional environmental monitoring, and land use censuses as part of their programs to assess and control dose within the proposed higher dose limits.

In addition to the proposed changes to allow requests for NRC authorization, on a case-by-case basis, for higher public dose limits in the controlled area and in the unrestricted area, the NRC is proposing two other changes to the requirements at § 20.1301: (1) changes to the limits that apply to caregivers of patients who cannot be released, and (2) deletion of the short-term dose rate limit for external sources in the unrestricted area.

First, the NRC is proposing to allow higher doses to members of the public who visit and/or care for medical patients who cannot be released under the provisions of § 35.75. Limits that apply to patient release are contained in 10 CFR part 35. The NRC has long permitted a member of the public to receive up to 500 mrem at the Authorized User's (AU) discretion under § 20.1301(c). Additionally, the NRC has approved several exemptions to this regulation as described in Regulatory Issue Summary (RIS) 2006–18, “Requesting Exemption from the Public Dose Limits for Certain Caregivers of Hospital Patients.” These exemptions allowed a caregiver to receive up to 2 rem, by default, with the flexibility to increase that amount if it was too low for a particular case. Similar to these previously issued exemptions, the NRC proposes to revise § 20.1301(c) to include a 2-rem limit for a caregiver and a 500-mrem limit for a non-caregiver member of the public per administration regimen. These limits are independent of the patient release regulations in 10 CFR 35.75, and doses accrued by caregivers or non-caregiver members of the public from exposure to a patient prior to release do not contribute to the patient's release evaluation. The guidance in RIS 2006–18 and the exemption process remain available to licensees should they desire to pursue authorization, on a case-by-case basis, to exceed the proposed new 2-rem limit for caregivers.

Amending the regulations to allow caregivers to receive up to 2 rem without a licensee requesting and having approved an exemption increases licensee flexibility, reduces regulatory cost and burden, and enables licensees to provide more timely care to patients while maintaining the public health and safety. The justification for the higher limit to the caregiver (*i.e.*, 2 rem instead of 100 mrem) is that it is beneficial, or possibly essential, to the wellbeing of the patient for caregivers to have access to the patient and may, therefore, be considered an element of the patient's medical treatment.

Caregivers are usually members of the patient's family or someone close to the family or the patient. Caregivers receive no financial compensation for the comfort or support that they provide a patient and knowingly consent to being exposed above the public dose limit.

Additionally, the higher dose limit is temporary and would not significantly impact the caregiver's lifetime fatal cancer risk. Similarly, the justification for the higher limit to the non-caregiver (*i.e.*, 500 mrem instead of 100 mrem) is that the higher dose limit is temporary and would not significantly impact the individual's lifetime fatal cancer risk. A non-caregiver being exposed in a situation relevant to § 20.1301(c) is likely also a family member or a friend of the patient, thus their access to the patient provides relief during medical treatment. In ICRP Publication 60—which provided the ICRP's first set of comprehensive recommendations after the recommended public dose limit was changed to its current value—the ICRP stated in paragraph 192 that, “Since the detriment is a function of the accumulation of dose over many years, it would be unduly restrictive to require the controls to be related rigidly to annual dose limits. Some flexibility in the limits is desirable.” Accordingly, with this proposed rulemaking the NRC is proposing to exercise this flexibility as it relates to patient care.

Second, the NRC is proposing to delete the short-term dose rate limit in the unrestricted area of 0.002 rem in any hour in § 20.1301(b) because it serves no safety purpose that is not already achieved by the public dose limit of 100 mrem per year. This is because a member of the public could receive the full 100 mrem annual public dose instantaneously and this fact in and of itself would have minimal safety impact. The actual concern in such a hypothetical case would be that a source that could provide such a high dose rate would most assuredly result in an exceedance of the annual public dose limit in a short period of time and it is

that exceedance that would require appropriate corrective actions and not the exceedance of any rate limit in receiving the dose. Licensees can voluntarily include short-term dose rate limits within their radiation protection programs for the purposes of dose management or to facilitate investigations of abnormal conditions; however, these are not necessary in addition to the annual public dose limit for the protection of the public health and safety. Therefore, the NRC is proposing to delete § 20.1301(a)(2) and the provision in § 20.1302(b)(2)(ii) that references 2 mrem in an hour.

License Termination Criteria

The NRC is proposing to revise the license termination criteria in 10 CFR part 20 subpart E to make conforming changes based on the discontinuation of the use of ALARA terminology in the NRC's regulations and guidance. Specifically, the NRC is removing the terms “as low as reasonably achievable” and “ALARA” from subpart E and replacing them, as appropriate. As explained previously in this document, the NRC's regulations and guidance would continue to use justification and optimization analyses to satisfy the original intent of the ALARA principle of, in this instance, reducing residual radioactivity to levels where further reductions would not be justified for both unrestricted and restricted use. The NRC intends for licensees to continue providing a combination of qualitative and quantitative analyses consistent with the guidance in NUREG–1757, volume 2, appendix N to satisfy the requirement to justify such reductions in residual radioactivity.

Additionally, the NRC is proposing to include a reference to § 20.1406(c) to its radiological criteria for unrestricted use at § 20.1402 to emphasize that the reduction in residual radioactivity may be justified by actions already taken by the licensee to minimize contamination. For example, licensees may be able to take credit for performing dismantlement and remediation activities in a way that minimizes the introduction of contamination to the environment in the analysis to demonstrate that further reductions are not justified. The NRC's proposed reference to this requirement would provide additional flexibility in the justification analysis required to comply with the proposed radiological criteria for unrestricted and restricted use.

With respect to restricted use, under § 20.1403 and § 20.1404, the NRC is proposing to increase clarity for licensees to demonstrate compliance with the proposed requirement that

further reductions in residual radioactivity necessary to comply with the provisions of § 20.1402 would not be justified. Currently, licensees must demonstrate that such reductions either would result in net public, or environmental harm or are consistent with ALARA. The NRC is proposing to introduce significantly more clarity by updating the requirement to include an explicit option to provide a cost-benefit analysis.

Finally, the NRC is making the language in 10 CFR part 20 subpart E consistent by using the broader term “traffic accidents” throughout instead of sometimes using the narrower term “deaths from transportation accidents.” The term “traffic accidents” encompasses the term “deaths from transportation accidents.”

Respiratory Protection

The NRC is proposing to make changes to its respiratory protection regulations to enable more efficient authorizations of respiratory equipment and their use. Currently, licensees are required to obtain individual authorization from the NRC in order to: (1) use respiratory equipment that has not been authorized by the National Institute for Occupational Safety and Health (NIOSH), and (2) use assigned protection factors (APFs) in excess of those specified in 10 CFR part 20. Through this rulemaking, a licensee would be allowed instead to reference approvals that the NRC has already issued to other licensees to use non-NIOSH certified equipment and/or to exceed the APFs in 10 CFR part 20; provided that the conditions in the safety evaluations used by the NRC to justify such approvals are applicable to that licensee. Effectively, this would mean that the NRC would only have to review and approve the use of new equipment or APFs once, and subsequent licensees could avail themselves of those approvals to the extent that they apply in their cases. This change would facilitate the use of modern equipment while minimizing risk and administrative burden because regulatory experience has shown that these reviews have not significantly differed once a precedent is established. In cases where the conditions of prior approvals would not apply or where licensee use of the equipment would not be within the scope of what the NRC considered in its safety evaluation, the licensee would have to individually apply for authorization as the regulations currently require.

Doses Received by a Member of the Public Due to Byproduct Material Administered for Medical Purposes

The NRC is proposing changes to 10 CFR part 35 regarding the release of patients who have been administered byproduct material. The first proposed change would add to 10 CFR 35.2 the definitions of “caregiver” and “administration regimen.” The proposed definition of “caregiver” is an adult who provides the patient with support or comfort for non-commercial gains following administration of byproduct material. The proposed definition of “administration regimen” is the course of administrations of a given radiopharmaceutical or brachytherapy source as intended by the authorized user. The second proposed change would amend 10 CFR 35.75 to add a provision to allow a consenting caregiver, who has been educated on the risks of radiation exposure, to receive up to 5 rem (*i.e.*, the occupation dose limit of 10 CFR 20.1201) per patient administration regimen. The dose limit to the general public would remain at 0.5 rem; however, the rule would be revised to specify that this limit is per administration regimen instead of per release. The third proposed change would amend 10 CFR 35.2075 to remove the requirement to keep records of dose evaluations for individual releases that meet certain criteria. Instead, the proposed change would require licensees to develop, implement, and maintain a written procedure for ensuring that a member of the public is not likely to exceed the established limits. The licensee would be required to retain this procedure for the duration of the license. Additionally, the proposed change would remove the requirement to document each instance where instructions are given to a breastfeeding patient.

In 2002, the NRC adopted the current dose-based criteria used in 10 CFR 35.75. At that time, the patient release framework was developed to address brachytherapy implants and single administration therapies (*e.g.*, I-131 therapy for conditions of the thyroid). Also at that time, other applications of byproduct material use in medicine did not involve quantities that would warrant concern regarding the release of patients. The medical landscape has evolved substantially since the current 10 CFR 35.75 was put in place. Namely, radiopharmaceuticals are increasingly being administered over a series of administrations as a matter of protocol, instead of all at once. With the proposed changes to 10 CFR 35.75, the NRC intends to adapt its patient release

framework to better address the evolving use of byproduct material in medicine. While the limit for dose received by an individual member of the public is 0.5 rem per patient administration regimen, the NRC is proposing to introduce flexibility through the identification of a caregiver, who is eligible to receive up to the occupational dose limit (*i.e.*, 5 rem) per patient administration regimen. Individuals identified as caregivers (*e.g.*, spouses, parents) are likely to receive the most dose as a result of a given patient release due to high duration or close contact activities, such as co-sleeping. Other members of the public are unlikely to receive a dose on the same order of magnitude as these potential caregivers. This is reflected in DG-8061 (the proposed revision 2 to Regulatory Guide (RG) 8.39) by using an assumed occupancy factor of 0.25 at 1 meter for a general member of the public (also referred to as a bystander) and an assumed occupancy factor of 1 at 1 meter for a caregiver in the tables provided. RG 8.39 contains additional information regarding patient-specific calculations for added flexibility, including lower occupancies for both caregivers and other members of the public.

The NRC is also proposing to change the recordkeeping requirements for patient release outlined in 10 CFR 35.2075 to better align with common practice. Most patients are likely to fall into certain categories that a licensee has previously determined to be compliant with release in accordance with 10 CFR 35.75. It is unnecessarily burdensome for licensees to retain the bases of release for individual patients who are released in such a way. As such, the NRC is proposing to instead require licensees to develop, implement, and maintain a written procedure for releasing patients in accordance with 10 CFR 35.75. The NRC is also proposing to remove the requirement to individually document when a breastfeeding patient has been given instructions following administration of byproduct material. The procedures required by the proposed 10 CFR 35.75 should detail the process for ensuring compliance for all patients who have been administered byproduct material, including situations where licensees provide instructions to patients who are breastfeeding. This change is intended to alleviate the burden associated with prescriptive recordkeeping requirements and to instead focus on the need to maintain a robust patient release program that enables the treatment of

patients while protecting members of the public.

For patients who cannot be released under 10 CFR 35.75, the NRC is proposing to amend 10 CFR 20.1301 to incorporate into that rule flexibilities introduced in RIS 2006–18. The proposed changes to 10 CFR 20.1301(c) would permit licensees, without having to apply for an exemption, to allow a caregiver, newly defined in the proposed changes to 10 CFR part 35, to receive up to 2 rem while providing care to a patient who cannot be released under 10 CFR 35.75. Any receipt of dose in excess of 2 rem would still require an application for an exemption and case-by-case prior approval by the NRC.

In 2002, the NRC amended 10 CFR part 20 to allow some visitors of patients who cannot be released under 10 CFR 35.75 to receive up to 500 mrem. At that time, the NRC acknowledged that, because visitors are often family members or close friends of patients, there is a substantial benefit that outweighs the risk of additional exposure. However, as the medical use of byproduct material has changed, it has come to the NRC's attention that a limit of 500 mrem is insufficient and overly cautious for some situations. While RIS 2006–18 does not set a maximum dose for which a licensee may apply for an exemption, 2 rem is the established initial dose limit for the outlined exemption. The NRC has already deemed a limit of 2 rem adequately protective for situations where a caregiver is necessary while a patient is hospitalized, provided that the licensee justifies the use of the exemption. The proposed rule would therefore eliminate the need for licensees to apply for an exemption for caregiver doses up to 2 rem. For caregiver doses in excess of 2 rem, licensees may still refer to RIS 2006–18. Note that caregiver doses received prior to the patient's release do not contribute to the 5 rem allowed by 10 CFR 35.75 following release. Finally, the proposed update to 10 CFR 20.2107 would require licensees to retain a record of the justification for a caregiver's dose for three years following the exposure.

Industrial Radiography

The NRC is proposing to remove from 10 CFR part 34 the definition of the term “ALARA” in § 34.3 and the prescriptive list of the radiation safety officer (RSO) responsibilities in § 34.42(c), which includes the term “ALARA.” Removing this term is consistent with the other changes being made as part of this rulemaking and removing the RSO responsibilities is consistent with other similar descriptions of requirements for

RSOs. Licensees would continue to be able to look to guidance in NUREG–1556, volume 2, for more detailed information regarding the responsibilities of an RSO for industrial radiography.

Uniform Waste Manifest Forms

The NRC is also proposing changes to 10 CFR part 20 appendix G to provide additional clarity and flexibility on how the information requested on the Uniform Waste Manifest forms (NRC Forms 540 and 540A, 541 and 541A, and 542 and 542A) could be provided. The proposed changes would clarify that the Manifest does not need to include these NRC forms themselves as long as the Manifest reflects the information requested on the applicable NRC forms. Other proposed changes would remove language specifying when NRC Form 540 must physically accompany a shipment and would instead reference Department of Transportation (DOT) regulations. This change would ensure that the NRC's regulations are consistent with the DOT's regulations, including potential future changes to the DOT's regulations. The remaining proposed changes would add clarity and correct minor grammatical errors.

Clarifying Changes to Appendix A to 10 CFR Part 40

The NRC is proposing a change to the introduction section of Appendix A to 10 CFR part 40. The purpose of this change would be to clarify that the use of the phrase “as low as is reasonably achievable” in Appendix A has the same meaning as in EPA's generally applicable standards in 40 CFR part 192, which is different than how NRC has traditionally used ALARA. In this context, the use of the phrase “as low as reasonably achievable” is focused on the technical practicability of corrective actions. The remaining proposed changes to Appendix A are consistent with the proposed changes to 10 CFR part 20 in that they would require licensees to manage dose within the applicable limits and would require that practicable dose reduction measures be taken. The proposed changes would remove any language that implies that further dose reduction is required. In this context, the phrase “as low as reasonably achievable” will remain in Appendix A, Criterion 5B(6) to conform to EPA's generally applicable standards in 40 CFR part 192.

Conforming Changes to 10 CFR Parts 50, 61, 71, and 72

This proposed rule would also include additional conforming changes

to align with the removal of references to ALARA. This includes the removal of a reference to ALARA in 10 CFR 50.66 and in 10 CFR 71.78. Additionally, removals of references to ALARA are proposed in 10 CFR part 72, specifically in §§ 72.3, 72.24, 72.44, 72.104, and 72.126. The changes proposed to § 72.44(d)(3) are conforming changes to align with the proposed changes to effluent reporting requirements in 10 CFR 50.34a, which are discussed in the previous section entitled “Effluents.”

Finally, the NRC is proposing to revise the public dose limits in 10 CFR part 61 to make conforming changes based on the discontinuation of the use of ALARA terminology in the NRC's regulations and guidance. Specifically, the NRC is proposing to remove the terms “as low as reasonably achievable” and “ALARA” from subpart C, “Performance Objectives.”

Conforming Changes to 10 CFR Part 53

The changes proposed in this rulemaking necessitate conforming changes to the recently issued 10 CFR part 53. In general, these changes serve two purposes. First, they acknowledge that licensees may potentially use dosimetry systems that provide results in terms of total effective dose, as opposed to total effective dose equivalent. Second, they provide the additional flexibility that is being included in the proposed changes to the NRC's overall radiation protection regulatory framework based on the agency's reconsideration of its use of the LNT model and its application of the ALARA principle.

The regulations at § 53.210, § 53.425, and § 53.530 are proposed to be revised to add reference to total effective dose to the units for dose-based criteria. Additionally, footnote 1 to § 53.210 is proposed to be revised to delete “TEDE” as the designation of the type of effective dose is not central to the purpose of the footnote, rather the magnitude of the dose is. The requirements at § 53.850 are proposed to be revised to change “limiting” to “controlling” to be consistent with the intent of effluent monitoring and control measures. Additionally, § 53.850(b)(2) is proposed to be revised to remove reference to the “Annual Radiological Environmental Operating and Radioactive Effluent Release Reports,” in lieu of a more generalized requirement for an effluent program to retain records and develop reporting criteria. This change conforms to changes being proposed in this rulemaking for radiological effluent monitoring and control through revisions to § 50.34a, § 50.36a, and

The NRC is proposing to assign these regulations Category B because the use of dosimetry methods is cross-jurisdictional and the uniformity of approved alternatives is necessary to ensure an orderly regulatory pattern for the use and regulation of material between all Agreement States and NRC jurisdictions.

Next, the NRC is proposing to change the compatibility category of 10 CFR 20.1301(d) from Category C to Category A. Currently, 10 CFR 20.1301(d) allows a licensee or applicant to apply for prior NRC authorization to operate in excess of the public dose limit (*i.e.*, 100 mrem) for an individual member of the public, up to 500 mrem, and specifies the information the entity should provide in this application. The proposed revision to 10 CFR 20.1301(d) would remove the 500 mrem upper limit. This change does not affect the ability for Agreement States to approve an optional upper public dose limit consistent with past practices. With the removal of that upper limit and the regulation tied to the public dose limit of 100 mrem, the NRC proposes designating the regulation Compatibility Category A, consistent with the designation for the public dose limit.

For the newly proposed 10 CFR 20.1205, which concerns a new process for allowing workers to receive occupational doses in excess of applicable annual limits, the NRC is

proposing to assign Compatibility Category A. As explained above, this optional process in the newly proposed 10 CFR 20.1205 would allow licensees to periodically exceed annual occupational dose limits for workers, provided that an adequate decision-making process is applied to support its use and that the overall dose is limited within specified multi-year average values. The proposed approach would allow for dose averaging across several years to demonstrate compliance with occupational dose limits. Similar to 10 CFR 20.1301(d) for the public dose limit, 10 CFR 20.1205 allows licensees flexibility with respect to the occupational dose limit. Thus, the NRC proposes designating 10 CFR 20.1205 as Category A, consistent with the designation for the occupational dose limit itself and the similar proposed designation for 10 CFR 20.1301(d) for the public dose limit flexibility. Moreover, in order to protect radiation workers' ability to work in different jurisdictions within the same calendar year when utilizing this provision, it is necessary for all jurisdictions to have the same flexibility built into their regulations.

With respect to 10 CFR part 35, the NRC is proposing to assign compatibility Category B to the new definitions for "Caregiver" and "Administration regimen" as well as the revised 10 CFR 35.75(b) regarding the

dose limits for patient release, which means that these requirements have cross-jurisdictional boundaries implications. Agreement States' requirements should be essentially identical to those of the NRC so that there are consistent standards for patient release requirements between the NRC and the Agreement States.

Since the 2002 adoption of the current dose-based criteria in 10 CFR 35.75, the treatment and release of patients with byproduct materials have undergone a number of significant changes with regard to the doses and administrative regimens of therapeutic radioisotopes used, treatment modalities, and the treatment and release of patient at regional treatment centers that require patients to travel in multiple jurisdictions. These changes require consistent standards to ensure equivalent levels of protection. For example, an increasing number of therapeutic radiopharmaceuticals are being delivered over the course of multiple administrations. Safe implementation of the proposed revision to 10 CFR 35.75(b) relies on consistent application of release considerations over the course of the entire administration regimen, which could be delivered in multiple jurisdictions.

The compatibility (A, B, C, D, and NRC) and adequacy (H&S) categories are designated in the following tables:

ADEQUACY AND COMPATIBILITY TABLE FOR 10 CFR PART 19

Section	Change	Subject	Compatibility	
			Existing	New
19.12	Amend	Instruction to workers	C	C.
19.13	Amend	Notifications and reports to individuals ..	C	C.

ADEQUACY AND COMPATIBILITY TABLE FOR 10 CFR PART 20

Section	Change	Subject	Compatibility	
			Existing	New
20.1003	Amend	Definition—Airborne radioactivity area ...	A	A.
20.1003	Delete	Definition—ALARA	A.	
20.1003	Amend	Definition—Annual limit on intake	A	A.
20.1003	Amend	Definition—Committed Dose Equivalent	A	A.
20.1003	New	Definition—Committed equivalent dose		A.
20.1003	Amend	Definition—Committed effective dose equivalent.	A	A.
20.1003	New	Definition—Committed effective dose		A.
20.1003	Amend	Definition—Derived air concentration ...	A	A.
20.1003	Amend	Definition—Dose or radiation dose	D	D.
20.1003	Amend	Definition—Dose equivalent	A	A.
20.1003	New	Definition—Dosimetry method (or system).		A.
20.1003	New	Definition—Effective dose		A.
20.1003	Amend	Definition—Effective dose equivalent	A	A.
20.1003	New	Definition—Equivalent Dose		A.
20.1003	New	Definition—Graded approach to dose management.		A.
20.1003	Amend	Definition—License	D	D.
20.1003	Amend	Definition—Nonstochastic effect	A	A.
20.1003	Amend	Definition—Planned special exposure ...	D	D.

ADEQUACY AND COMPATIBILITY TABLE FOR 10 CFR PART 20—Continued

Section	Change	Subject	Compatibility	
			Existing	New
20.1003	New	Definition—Planned occupational dose limit extension.		A.
20.1003	New	Definition—Radiation weighting factor		A.
20.1003	Amend	Definition—Stochastic effects	A	A.
20.1003	New	Definition—Total effective dose		A.
20.1003	Amend	Definition—Weighting factor	A	A.
20.1004	Amend	Units of radiation dose	A	A.
20.1010	New	Dosimetry methods		B.
20.1101(a) and (c)	Amend	Radiation protection programs	H&S	H&S.
20.1101(b)	Amend	Radiation protection programs	H&S	A.
20.1101(d)	Amend	Radiation protection programs	C	A, except portions of these provisions are designated as NRC (Those portions of the provision that address areas reserved to the NRC, <i>e.g.</i> , 10 CFR Parts 50.34a and 53.260 are designated as a Compatibility Category NRC).
20.1201	Amend	Occupational dose limits for adults	A	A.
20.1202	Amend	Compliance with requirements for summation of external and internal doses.	A	A.
20.1204	Amend	Determination of internal exposure	A	A.
20.1205	New	Planned occupational dose limit extension.		A.
20.1206	Amend	Planned special exposures	D	D.
20.1301(a), (b), (c)	Amend	Dose limits for individual members of the public.	A	A.
20.1301(d)	Amend	Dose limits for individual members of the public.	C	A.
20.1302(b)	Amend	Compliance with dose limits for individual members of the public.	H&S	H&S.
20.1302(c)	Amend	Compliance with dose limits for individual members of the public.	D	D.
20.1402	Amend	Radiological criteria for unrestricted use	C	B—for source material recovered from any mineral resources processed primarily for purposes other than obtaining the source material content. C—for all other Agreement State licensed activities.*
20.1403	Amend	Criteria for license termination under restricted conditions.	C	B—for source material recovered from any mineral resources processed primarily for purposes other than obtaining the source material content. C—for all other Agreement State licensed activities.*
20.1404	Amend	Alternate criteria for license termination	C	B—for source material recovered from any mineral resources processed primarily for purposes other than obtaining the source material content. C—for all other Agreement State licensed activities.*
20.1405	Amend	Public notification and public participation.	C	C.
20.1502	Amend	Conditions requiring individual monitoring of external and internal occupational dose.	H&S	H&S.
20.1601(f)	Amend	Control of access to high radiation areas.	H&S	H&S.
20.1702	Amend	Use of other controls	H&S	H&S.
20.1703	Amend	Use of individual respiratory protection equipment.	H&S	H&S.
20.1704	Amend	Further restrictions on the use of respiratory protection equipment.	D	D.
20.1705	Amend	Application for use of higher assigned protection factors.	B	B.
20.1905(b)	Amend	Exceptions to labeling requirements	A	A.
20.2002	Amend	Method for obtaining approval of proposed disposal procedures.	D	D.
20.2003 (a)(2)&(a)(3)	Amend	Disposal by release into sanitary sewerage.	A	A.
20.2004	Amend	Treatment or disposal by incineration	D	D.
20.2101	Amend	General provisions	C	C.
20.2104	Amend	Determination of prior occupational dose.	D or H&S (for States who adopt planned special exposure).	D or H&S (for States who adopt planned special exposure).
20.2105	Amend	Records of planned special exposures	D	D.
20.2106(a)	Amend	Records of individual monitoring results	C	C.
20.2107	Amend	Records of dose to individual members of the public.	D	D.
20.2202(e)	Amend	Notification of incidents	D	D.

ADEQUACY AND COMPATIBILITY TABLE FOR 10 CFR PART 20—Continued

Section	Change	Subject	Compatibility	
			Existing	New
20.2203(a), (b)	Amend	Reports of exposures, radiation levels, and concentrations of radioactive material exceeding the constraints or limits.	C	C.
Appendix G	Amend	Requirements for Low-level radioactive waste intended for disposal at land disposal facilities and manifests.	B	B.
App. G I	Amend	Manifest	B	B.
App. G I	New	Definition—Carrier		B.
App. G I	Delete	Definition—Computer readable medium	B.	
App. G I	Amend	Definition—EPA identification number	B	B.
App. G I	Amend	Definition—High integrity container	B	B.
App. G I	Amend	Definition—NRC Forms 540, 540A, 541, 541A, 542, and 542A.	B	B.
App. G I	Amend	Definition—Shipping paper	B	B.
App. G I	Amend	Definition—Uniform Low-Level Radioactive Waste Manifest.	B	B.
App. G III.A	Amend	Control and Tracking	B	B.
App. G III.B	Amend	Control and Tracking	B	B.
App. G III.C	Amend	Control and Tracking	B	B.
App. G III.D	Amend	Control and Tracking	B	B.
Appendix H	New	Alternative Dosimetry Methods Acceptable for Use to Demonstrate Compliance with NRC's Standards for Protection Against Radiation.		B.

* Consistent with 10 CFR 20.1401(a), the criteria in 10 CFR part 20, subpart E do not apply to uranium and thorium recovery facilities already subject to appendix A to 10 CFR part 40 or the uranium solution extraction facilities.

ADEQUACY AND COMPATIBILITY TABLE FOR 10 CFR PART 34

Section	Change	Subject	Compatibility	
			Existing	New
34.3	Delete	Definitions—ALARA	[A].	
34.42(c), (d)	Delete	Radiation Safety Officer for industrial radiography.	D.	

ADEQUACY AND COMPATIBILITY TABLE FOR 10 CFR PART 35

Section	Change	Subject	Compatibility	
			Existing	New
35.2	New	Definitions—Administration Regimen		B.
35.2	New	Definitions—Caregiver		B.
35.75(a)	Revised	Release of individuals containing unsealed byproduct material or implants containing byproduct material.	C	C.
35.75(b)	Amend	Release of individuals containing unsealed byproduct material or implants containing byproduct material.	C	B.
35.75(c)	Amend	Release of individuals containing unsealed byproduct material or implants containing byproduct material.	C	C.
35.75(d)	Amend	Release of individuals containing unsealed byproduct material or implants containing byproduct material.	D	D.
35.2075	Amend	Records of the release of individuals containing unsealed byproduct material or implants containing byproduct material.	D	D.

§ 50.34a or § 53.260 of this chapter, such that the individual member of the public likely to receive the highest dose will not be expected to receive a total effective dose equivalent in excess of 25 mrem (0.25 mSv) per year from these emissions. If a licensee subject to this requirement exceeds this dose constraint, the licensee shall report the exceedance as provided in § 20.2203 and promptly take appropriate corrective action to ensure against recurrence. A licensee or applicant may request prior NRC authorization to establish a higher constraint, provided that the proposed constraint provides an ample margin of safety to protect public health.

■ 9. Revise and republish § 20.1201(a) through (d) to read as follows:

§ 20.1201 Occupational dose limits for adults.

(a) The licensee shall control the occupational dose to individual adults, except for planned occupational dose limit extensions under § 20.1205 and planned special exposures under § 20.1206, to the following dose limits.

(1) An annual limit, which is the more limiting of—

(i) The total effective dose equivalent being equal to 5 rems (0.05 Sv); or

(ii) The sum of the deep-dose equivalent and the committed dose equivalent to any individual organ or tissue other than the lens of the eye being equal to 50 rems (0.5 Sv).

(2) The annual limits to the lens of the eye, to the skin of the whole body, and to the skin of the extremities, which are:

(i) A lens dose equivalent of 15 rems (0.15 Sv), and

(ii) A shallow-dose equivalent of 50 rem (0.5 Sv) to the skin of the whole body or to the skin of any extremity.

(b) Doses received in excess of the annual limits, with the exception of doses received under § 20.1205, including doses received during accidents, emergencies, and planned special exposures, must be subtracted from the limits for planned special exposures that the individual may receive during the current year (see § 20.1206(e)(1)) and during the individual's lifetime (see § 20.1206(e)(2)).

(c) When the external exposure is determined by measurement with an external personal monitoring device, the deep-dose equivalent must be used in place of the effective dose equivalent, unless the effective dose equivalent is determined by a dosimetry method approved by the NRC or by an alternative dosimetry method pursuant to § 20.1010. The assigned deep-dose equivalent must be for the part of the

body receiving the highest exposure. The assigned shallow-dose equivalent must be the dose averaged over the contiguous 10 square centimeters of skin receiving the highest exposure. The deep-dose equivalent, lens-dose equivalent, and shallow-dose equivalent may be assessed from surveys or other radiation measurements for the purpose of demonstrating compliance with the occupational dose limits, if the individual monitoring device was not in the region of highest potential exposure, or the results of individual monitoring are unavailable.

(d) Derived air concentration (DAC) and annual limit on intake (ALI) values are presented in table 1 of appendix B to part 20 and may be used to determine the individual's dose (see § 20.2106) and to demonstrate compliance with the occupational dose limits. Alternatively, the licensee may use alternative dosimetry methods to demonstrate compliance with the occupational dose limits (see § 20.1010).

* * * * *

■ 10. In § 20.1202, revise footnote 1 to read as follows:

§ 20.1202 Compliance with requirements for summation of external and internal doses.

* * * * *

^[1] An organ or tissue is deemed to be significantly irradiated if, for that organ or tissue, the product of the weighting factor, w_T , and the committed dose equivalent, $H_{T,Q,50}$, per unit intake is greater than 10 percent of the maximum weighted value of $H_{T,Q,50}$, (i.e., $w_T H_{T,Q,50}$) per unit intake for any organ or tissue.

■ 11. In § 20.1204, revise paragraphs (c)(2) and (3), (e)(1), and (h)(2), to read as follows:

§ 20.1204 Determination of internal exposure.

* * * * *

(c) * * *

(2) Upon prior approval of the Commission or consistent with the use of alternative dosimetry methods to demonstrate compliance with dose limits as allowed by § 20.1010, adjust the DAC or ALI values to reflect the actual physical and chemical characteristics of airborne radioactive material (e.g., aerosol size distribution or density); and

(3) If applicable, separately assess the contribution of fractional intakes of different solubility classes (e.g., Class D, W, or Y compounds of a given radionuclide (see appendix B to part 20)) to the committed effective dose equivalent.

* * * * *

(e) * * *

(1) The sum of the ratios of the concentration to the appropriate DAC value (e.g., D, W, Y) from appendix B to part 20, or as determined using an alternative dosimetry method as allowed by § 20.1010, for each radionuclide in the mixture; or

* * * * *

(h) * * *

(2) When the ALI (and the associated DAC) is determined by the nonstochastic organ dose limit of 50 rem (0.5 Sv), the licensee may, as a simplifying assumption, use the stochastic ALIs to determine committed effective dose equivalent. However, if the licensee uses the stochastic ALIs, the licensee must also demonstrate that the limit in § 20.1201(a)(1)(ii) is met. Table 1 of appendix B to part 20 lists stochastic ALIs, even when the limiting ALI and DAC is governed by the nonstochastic organ dose limit of 50 rem (0.5 Sv). In cases where a licensee is using an alternative dosimetry method pursuant to § 20.1010, that licensee can develop the stochastic ALI to use for the purposes of this paragraph.

■ 12. Add section § 20.1205 to subpart C to read as follows:

§ 20.1205 Planned occupational dose limit extension.

A licensee may authorize a worker that is not a declared pregnant woman or a minor to receive occupational doses in excess of the limits specified in § 20.1201 provided that each of the following conditions is satisfied—

(a) The licensee (and employer if the employer is not the licensee) authorizes the planned occupational dose limit extension, in writing, before the dose limit specified in § 20.1201 is exceeded.

(b) Before a planned occupational dose limit extension, the licensee ensures that the individuals involved are—

(1) Informed of the purpose of the planned operation;

(2) Informed of the estimated doses and associated potential risks and estimated radiation levels or other conditions that might be involved in performing the task; and

(3) Instructed in the measures to be taken to manage doses considering other risks that may be present.

(c) Prior to permitting an individual to participate in a planned occupational dose limit extension, the licensee ascertains prior occupational doses as required by § 20.2104(b) during the current year and the preceding four years for each individual involved.

(d) The licensee does not authorize a planned occupational dose limit extension that would cause an

individual to receive a dose from all occupational exposures in excess of—

(1) Five times the numerical limits in § 20.1201(a)(1) and § 20.1201(a)(2)(ii) over the current year and the preceding four years;

(2) Twice the numerical limits in § 20.1201(a)(1) and § 20.1201(a)(2)(ii) in any one year; and

(3) A lens dose equivalent of 15 rem (0.15 Sv) in any one year.

(e) The licensee maintains records of the conduct of a planned occupational dose limit extension in accordance with § 20.2105.

(f) The dose from planned occupational dose limit extensions is not to be considered in controlling future occupational dose of the individual under § 20.1201(a) but is to be included in evaluations required by §§ 20.1205 (c) and (d) and 20.1206 (d) and (e).

■ 13. In § 20.1206, revise paragraph (c)(3) to read as follows:

§ 20.1206 Planned special exposures.

* * * * *

(c) * * *

(3) Instructed in the measures to be taken to manage doses considering other risks that may be present.

* * * * *

■ 14. In § 20.1301, revise paragraph (a)(1), remove and reserve paragraph (a)(2), and revise paragraphs (b) through (d) to read as follows:

§ 20.1301 Dose limits for individual members of the public.

(a) * * *

(1) The total effective dose equivalent to individual members of the public from the licensed operation, or any other source of radiation under the control of a licensee, does not exceed 0.1 rem (1 mSv) in a year, exclusive of the dose contributions from background radiation, from any medical administration the individual has received, from exposure to individuals administered radioactive material and released under § 35.75, from voluntary participation in medical research programs, and from the licensee's disposal of radioactive material into sanitary sewerage in accordance with § 20.2003.

(2) [Reserved]

(b) If the licensee permits members of the public to have access to controlled areas, the limits for members of the public continue to apply to those individuals. Alternatively, a licensee or applicant may request prior NRC authorization for an annual dose limit in excess of 0.1 rem (1 mSv) for members of the public who have access to controlled areas, provided that dose is appropriately managed.

(c) Notwithstanding paragraph (a)(1) of this section, a licensee may permit members of the public to visit an individual who cannot be released, under § 35.75, to receive a radiation dose in excess of 0.1 rem (1 mSv) if—

(1) The radiation dose received is not likely to exceed either—

(i) 0.5 rem (5 mSv) for a member of the public who is not a caregiver per administration regimen; or

(ii) 2 rem (20 mSv) for a caregiver, as defined in 10 CFR part 35, per administration regimen; and

(2) The authorized user, as defined in 10 CFR part 35, has determined before the visit that it is appropriate.

(d) A licensee or applicant may apply for prior NRC authorization to operate with an annual dose limit for an individual member of the public in excess of 0.1 rem (1 mSv). The licensee or applicant shall include the following information in this application:

(1) Demonstration of the need for and the expected duration of operations in excess of the limit in paragraph (a) of this section;

(2) Description of the licensee's program to assess and control dose within the proposed dose limit for an individual member of the public; and

(3) The proposed dose limit for an individual member of the public and its supporting basis, including why it remains protective of the public health and safety.

* * * * *

■ 15. In § 20.1302, revise paragraph (b)(2) and remove and reserve paragraph (c) to read as follows:

§ 20.1302 Compliance with dose limits for individual members of the public.

* * * * *

(b) * * *

(1) * * *

(2) Demonstrating that—

(i) The annual average concentrations of radioactive material released in gaseous and liquid effluents at the boundary of the unrestricted area do not exceed the values specified in table 2 of appendix B to part 20 or as developed using alternative dosimetry methods pursuant to § 20.1010; and

(ii) If an individual were continuously present in an unrestricted area, the dose from external sources would not exceed 0.05 rem (0.5 mSv) in a year.

(c) [Reserved]

■ 16. Revise § 20.1402 to read as follows:

§ 20.1402 Radiological criteria for unrestricted use.

A site will be considered acceptable for unrestricted use if the residual radioactivity that is distinguishable

from background radiation results in a TEDE to the average member of the critical group that does not exceed 25 mrem (0.25 mSv) per year, including that from groundwater sources of drinking water, and, consistent with § 20.1406(c), the residual radioactivity has been reduced to levels where further reductions would not be justified when considering any detriments, such as traffic accidents, expected to potentially result from decontamination and waste disposal.

■ 17. In § 20.1403, revise paragraph (a) and revise and republish paragraph (e) to read as follows:

§ 20.1403 Criteria for license termination under restricted conditions.

* * * * *

(a) The licensee can demonstrate that further reductions in residual radioactivity necessary to comply with the provisions of § 20.1402 would not be justified when considering any detriments, such as traffic accidents, expected to potentially result from decontamination and waste disposal; would result in net public or environmental harm; or would not be justified through a cost-benefit analysis;

* * * * *

(e) Residual radioactivity at the site has been reduced so that if the institutional controls were no longer in effect, there is reasonable assurance that the TEDE from residual radioactivity distinguishable from background to the average member of the critical group would not exceed either—

(1) 100 mrem (1 mSv) per year provided that the licensee—

Demonstrates that further reductions in residual radioactivity would not be justified when considering any detriments, such as traffic accidents, expected to potentially result from decontamination and waste disposal; or

(2) 500 mrem (5 mSv) per year provided that the licensee—

(i) Demonstrates that further reductions in residual radioactivity necessary to comply with the 100 mrem/y (1 mSv/y) value of paragraph (e)(1) of this section are not technically achievable, would be prohibitively expensive, or would result in net public or environmental harm;

(ii) Makes provisions for durable institutional controls;

(iii) Provides sufficient financial assurance to enable a responsible government entity or independent third party, including a governmental custodian of a site, both to carry out periodic rechecks of the site no less frequently than every 5 years to assure that the institutional controls remain in place as necessary to meet the criteria of

§ 20.1403(b) and to assume and carry out responsibilities for any necessary control and maintenance of those controls. Acceptable financial assurance mechanisms are those in paragraph (c) of this section.

■ 18. In § 20.1404, revise and republish paragraphs (a)(1) through (a)(4) to read as follows:

§ 20.1404 Alternate criteria for license termination.

(a) * * *

(1) Provides assurance that public health and safety would continue to be protected by submitting an analysis demonstrating that it is unlikely that the dose from all man-made sources combined, other than medical, would be more than 100 mrem/y (1 mSv/y);

(2) Has employed to the extent practical restrictions on site use according to the provisions of § 20.1403 in minimizing exposures at the site;

(3) Demonstrates that further reductions in residual radioactivity would not be justified when considering any detriments, such as traffic accidents, expected to potentially result from decontamination and waste disposal;

(4) Has submitted a decommissioning plan or License Termination Plan (LTP) to the Commission indicating the licensee's intent to decommission in accordance with §§ 30.36(d), 40.42(d), 50.82(a) and (b), subpart G of part 53, 70.38(d), or 72.54 of this chapter, and specifying that the licensee proposes to decommission by use of alternate criteria. The licensee shall document in the decommissioning plan or LTP how the advice of individuals and institutions in the community who may be affected by the decommissioning has been sought and addressed, as appropriate, following analysis of that advice. In seeking such advice, the licensee shall provide for:

(i) Participation by representatives of a broad cross section of community interests who may be affected by the decommissioning;

(ii) An opportunity for a comprehensive, collective discussion on the issues by the participants represented; and

(iii) A publicly available summary of the results of all such discussions, including a description of the individual viewpoints of the participants on the issues and the extent of agreement and disagreement on the issues; and

* * * * *

■ 19. In § 20.1405, revise paragraph (b) to read as follows:

§ 20.1405 Public notification and public participation.

* * * * *

(b) Publish a notice in the **Federal Register** and in a forum, such as local newspapers, letters to State or local organizations, or other appropriate forum, that is readily accessible to individuals in the vicinity of the site, and solicit comments from affected parties.

■ 20. In § 20.1502, revise introductory text and paragraphs (a)(4) and (b)(1) to read as follows:

§ 20.1502 Conditions requiring individual monitoring of external and internal occupational dose.

Each licensee shall monitor exposures to radiation and radioactive material at levels sufficient to demonstrate compliance with the occupational dose limits of this part.

(a) * * *

(4) Individuals entering a high or very high radiation area. These individuals shall be monitored for external occupational dose, unless any of the conditions of § 20.1502(b) apply, in which case they shall be monitored for both external and internal occupational dose.

(b) * * *

(1) Adults likely to receive, in 1 year, an intake in excess of 10 percent of the applicable ALI(s) in table 1, columns 1 and 2, of appendix B to §§ 20.1001–20.2402 or as developed using alternative dosimetry methods pursuant to § 20.1010;

* * * * *

§ 20.1601 [Amended]

■ 21. In § 20.1601(f), remove the text “ALARA”.

■ 22. Revise and republish § 20.1702 to read as follows:

§ 20.1702 Use of other controls.

(a) When it is not practical to apply process or other engineering controls to control the concentrations of radioactive material in the air to values below those that define an airborne radioactivity area, the licensee shall increase monitoring and limit intakes by one or more of the following means—

- (1) Control of access;
- (2) Limitation of exposure times;
- (3) Use of respiratory protection

equipment; or

- (4) Other controls.

(b) If the licensee performs an analysis to determine whether or not respirators should be used, the licensee may consider safety factors other than radiological factors. The licensee should also consider the impact of respirator use on workers' industrial health and safety.

■ 23. In § 20.1703, revise paragraph (b) to read as follows:

§ 20.1703 Use of individual respiratory protection equipment.

* * * * *

(b) If the licensee wishes to use equipment that has not been tested or certified by NIOSH, or for which there is no schedule for testing or certification, the licensee shall submit an application to the NRC for authorized use of this equipment except as provided in this part. The application must include evidence that the material and performance characteristics of the equipment are capable of providing the proposed degree of protection under anticipated conditions of use. This must be demonstrated either by licensee testing or on the basis of reliable test information. The use of equipment that has not been tested or certified by NIOSH, but has previously been approved for use by the NRC, does not require an application and its approval as described in this paragraph, provided that the licensee maintains an evaluation to demonstrate that the bases for the previous NRC approval—as documented in the applicable safety evaluation—are applicable to the licensee's facility.

* * * * *

§ 20.1704 [Amended]

■ 24. In § 20.1704(a), remove the text “ALARA” and add in its place, “within the requirements of this part”.

■ 25. In § 20.1705, add paragraph (c) to read as follows:

§ 20.1705 Application for use of higher assigned protection factors.

* * * * *

(c) The use of higher assigned protection factors that have previously been approved for use by the Commission does not require an application and its approval as described in this section, provided that the licensee maintains an evaluation to demonstrate that the bases for the previous Commission approval—as documented in the applicable safety evaluation—are applicable to the licensee's facility.

■ 26. In § 20.1905, revise paragraph (b) to read as follows:

§ 20.1905 Exemptions to labeling requirements.

* * * * *

(b) Containers holding licensed material in concentrations less than those specified in table 3 of appendix B to part 20 or concentrations derived using alternative dosimetry methods pursuant to § 20.1010; or

* * * * *

§ 20.2002 [Amended]

■ 27. In § 20.2002(d), remove the text “ALARA and”.

■ 28. In § 20.2003, revise paragraphs (a)(2) and (a)(3)(i) to read as follows:

§ 20.2003 Disposal by release into sanitary sewerage.

(a) * * *

(1) * * *

(2) The quantity of licensed or other radioactive material that the licensee releases into the sewer in 1 month divided by the average monthly volume of water released into the sewer by the licensee does not exceed the concentration listed in table 3 of appendix B to part 20 or concentrations derived using alternative dosimetry methods pursuant to § 20.1010; and

(3) * * *

(i) The licensee shall determine the fraction of the limiting concentration in table 3 of appendix B to part 20, or the limiting concentrations derived using alternative dosimetry methods pursuant to § 20.1010, represented by discharges into sanitary sewerage by dividing the actual monthly average concentration of each radionuclide released by the licensee into the sewer by the applicable limiting concentration; and

* * * * *

§ 20.2004 [Amended]

■ 29. In § 20.2004, remove from the first sentence in paragraph (b)(1) the text “the requirements of appendix I to part 50 of this chapter and”.

§ 20.2101 [Amended]

■ 30. In § 20.2101, remove from paragraph (c) the text “Notwithstanding” and replace with “Notwithstanding”.

■ 31. In § 20.2104, revise paragraphs (b) and (e)(2) to read as follows:

§ 20.2104 Determination of prior occupational dose.

* * * * *

(b) Prior to permitting an individual to participate in a planned occupational dose limit extension or a planned special exposure, the licensee shall determine—

(1) For a planned occupational dose limit extension—

(i) Prior occupational doses during the current year and the preceding four years that correspond to the applicable limits in § 20.1205(d) for each individual involved.

(2) For a planned special exposure—

(i) The internal and external doses from all previous planned special exposures; and

(ii) All doses in excess of the limits (including doses received during

accidents and emergencies) received during the lifetime of the individual.

* * * * *

(e) * * *

(2) That the individual is not available for planned occupational dose limit extensions, if records for the current year and the preceding four years are not complete, and planned special exposures.

* * * * *

■ 32. In § 20.2105:

■ a. Revise the section heading and paragraphs (a)(5) and (6);

■ b. Redesignate paragraph (b) as paragraph (c); and

■ c. Add a new paragraph (b).

The revisions and addition read as follows:

§ 20.2105 Records of planned occupational dose limit extensions and planned special exposures.

(a) * * *

(1) * * *

(5) How doses were managed; and

(6) The doses received by individuals involved in the planned special exposure.

(b) For each use of the provisions of § 20.1205 for planned occupational dose limit extensions, the licensee shall maintain records that describe the circumstances requiring the extension of occupational dose limits, how doses were managed, and the doses received by individuals involved in the planned occupational dose limit extension.

■ 33. In § 20.2106, revise paragraph (a) introductory text and paragraphs (a)(5) through (7) to read as follows:

§ 20.2106 Records of individual monitoring results.

(a) *Recordkeeping requirement.* Each licensee shall maintain records of doses received by individuals for whom monitoring was required pursuant to § 20.1502 and records of doses received during planned occupational dose limit extensions, planned special exposures, accidents, and emergency conditions. These records^[5] must include, when applicable—

(1) * * *

(5) The total effective dose equivalent when required by § 20.1202;

(6) The total of the deep-dose equivalent and the committed dose to the organ receiving the highest total dose; and

(7) In cases where monitoring was required pursuant to § 20.1502, but the dose received did not exceed 10 percent of the applicable monitoring criteria, the licensee may, instead of recording the numerical value of the dose received, annotate that an occupational dose was

received but did not exceed the criteria for recording.

* * * * *

^[5] Assessments of dose equivalent and records made using units in effect before the licensee's adoption of this part need not be changed.

■ 34. In § 20.2107, add paragraphs (c) and (d) to read as follows:

§ 20.2107 Records of dose to individual members of the public.

* * * * *

(c) The licensee shall retain a record of the justification for the dose received by a caregiver as described in § 20.1301(c)(1)(ii).

(d) The licensee shall retain the records required by paragraph (c) of this section for 3 years after the final date of the allowed exposure.

■ 35. In § 20.2202, revise paragraph (e) to read as follows:

§ 20.2202 Notification of incidents.

* * * * *

(e) The provisions of this section do not include doses that result from planned occupational dose limit extensions or from planned special exposures that are within the limits for planned special exposures, and that are reported under § 20.2204.

■ 36. In § 20.2203:

■ a. revise and republish paragraph (a)(2) and

■ b. remove the text “ALARA” from paragraph (b)(iv).

The revision reads as follows:

§ 20.2203 Reports of exposures, radiation levels, and concentrations of radioactive material exceeding the constraints or limits.

(a) * * *

(1) * * *

(2) Doses in excess of any of the following:

(i) The occupational dose limits for adults in § 20.1201 unless they are exceeded pursuant to § 20.1205 or § 20.1206; except that reporting of unplanned exceedances of the limit in 20.1201(a)(1)(i) is required only if the total effective dose equivalent for the current year and the preceding four years exceeds 25 rem; or

(ii) The occupational dose limits for a minor in § 20.1207; or

(iii) The limits for an embryo/fetus of a declared pregnant woman in § 20.1208; or

(iv) The limits for an individual member of the public in § 20.1301, except that reporting of exceedances of the limit in § 20.1301(a) is required only if the total dose to a single individual for the current year and the preceding four years exceeds 500 mrem; or

(v) Any applicable limit in the license; or

- (vi) The constraints for air emissions established under § 20.1101(d); or
- (vii) The applicable dose limit for a planned occupational dose limit extension in § 20.1205(c); or
- (viii) The applicable dose limit for a planned special exposure in § 20.1206(e); or

* * * * *

- 37. In appendix G to 10 CFR part 20:
 - a. Revise and republish the introductory paragraphs to Section I; add a definition for “Carrier”; remove the definition for “Computer readable medium”; and revise the definitions for “EPA identification number”, “High integrity container”, “NRC Forms 540, 540A, 541, 541A, 542, and 542A”, “Shipping paper”, and “Uniform Low-Level Radioactive Waste Manifest or uniform manifest”;
 - b. Revise paragraph III.A.6, paragraph III.B.4, paragraph III.C.7, and paragraph III.D.2.; and
 - c. Amend paragraph III.E.1. to remove the text “or receipt” and add, in its place, “of receipt”.

The revisions read as follows:

Appendix G to Part 20—Requirements for Transfers of Low-Level Radioactive Waste Intended for Disposal at Licensed Land Disposal Facilities and Manifests

I. Manifest

A waste generator, collector, or processor who transports, or offers for transportation, low-level radioactive waste intended for ultimate disposal at a licensed low-level radioactive waste land disposal facility must prepare a Manifest (OMB Control Numbers 3150-0164, -0165, and -0166) reflecting information requested on applicable NRC Forms 540 (Uniform Low-Level Radioactive Waste Manifest (Shipping Paper)) and 541 (Uniform Low-Level Radioactive Waste Manifest (Container and Waste Description)) and, if necessary, on an applicable NRC Form 542 (Uniform Low-Level Radioactive Waste Manifest (Manifest Index and Regional Compact Tabulation)). The Manifest does not need to use these NRC Forms themselves as long as the Manifest reflects the information requested on the applicable NRC Forms. References to NRC Forms 540, 540A, 541, 541A, 542, and 542A in this appendix refer to either the NRC Form(s) or any other document(s) reflecting the information requested on the NRC Form(s) (e.g., licensee-generated versions of the NRC Forms).

Upon agreement between shipper and consignee, NRC Forms 540 and 540A, 541 and 541A, and 542 and 542A may be completed, signed, transmitted, and stored in electronic media with the capability for producing legible, accurate, and complete records on the respective forms. NRC Forms 540 and 540A must be transmitted to the carrier in accordance with regulations of the Department of Transportation (DOT).

Licensees are not required by NRC to comply with the manifesting requirements of this part when they ship:

- (a) LLW for processing and expect its return (i.e., for storage under their license) prior to disposal at a licensed land disposal facility;

- (b) LLW that is being returned to the licensee who is the “waste generator” or “generator,” as defined in this part; or

- (c) Radioactively contaminated material to a “waste processor” that becomes the processor’s “residual waste.”

For guidance in completing these forms, refer to the instructions that accompany the forms. Copies of manifests required by this appendix may be legible carbon copies, photocopies, computer printouts, or electronic copies that reproduce the data of the uniform manifest.

NRC Forms 540, 540A, 541, 541A, 542 and 542A, and the accompanying instructions, in hard copy, may be obtained by writing or calling the Office of the Chief Information Officer, U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, telephone (301) 415-5877, or by visiting the NRC’s website at <http://www.nrc.gov> and selecting forms from the index found on the home page.

This appendix includes information requirements of the DOT, as codified in 49 CFR parts 172, 174, 175, 176, and 177. Information on hazardous, medical, or other waste, required to meet Environmental Protection Agency (EPA) regulations, as codified in 40 CFR parts 259, 261, or elsewhere, is not addressed in this section. However, any forms required by the EPA must accompany the Uniform Low-Level Radioactive Waste Manifest required by this chapter.

As used in this appendix, the following definitions apply:

Carrier means a person who transports passengers or property in commerce by rail car, aircraft, motor vehicle, or vessel.

* * * * *

EPA identification number means the number received by a carrier following application to the Administrator of EPA as required by 40 CFR part 263.

* * * * *

High integrity container (HIC) means a container commonly designed to meet the structural stability requirements of § 61.56 of this chapter, and to meet DOT requirements for a Type A package.

* * * * *

NRC Forms 540, 540A, 541, 541A, 542, and 542A mean either the NRC Form(s) or any other document(s) reflecting the information requested on the NRC Form(s) (e.g., licensee-generated or Agreement State-generated versions of the NRC Forms).

* * * * *

Shipping paper means NRC Form 540 and, if required, NRC Form 540A which includes the information required by DOT regulations.

* * * * *

Uniform Low-Level Radioactive Waste Manifest or uniform manifest means the combination of NRC Forms 540, 541, and, if necessary, 542, and their respective continuation sheets as needed.

* * * * *

III. Control and Tracking

A. * * *

- 6. If required by regulations of the DOT, physically include NRC Form 540 (and NRC Form 540A, if required) with the shipment regardless of the option chosen in paragraph A.5 of this section;

* * * * *

B. * * *

- 4. If required by regulations of the DOT, physically include NRC Form 540 (and NRC Form 540A, if required) with the shipment regardless of the option chosen in paragraph B.3 of this section;

* * * * *

C. * * *

- 7. If required by regulations of the DOT, physically include NRC Form 540 (and NRC Form 540A, if required) with the shipment regardless of the option chosen in paragraph C.6 of this section;

* * * * *

D. * * *

- 2. Maintain copies of or electronically store all completed manifests and electronically store the information required by 10 CFR 61.80(l) until the Commission terminates the license; and

* * * * *

- 38. Add new appendix H to part 20 to read as follows:

Appendix H to Part 20—Alternative Dosimetry Methods Acceptable for Use To Demonstrate Compliance With NRC’s Standards for Protection Against Radiation

(a) *Material incorporated by reference.* The material listed in this appendix for developing alternative dosimetry methods is incorporated by reference into § 20.1010 with the approval of the Director of the Office of the Federal Register under 5 U.S.C. 552(a) 1 CFR part 51. All approved material is available for inspection at the Nuclear Regulatory Commission (NRC) and at the National Archives Records Administration (NARA). Contact the NRC at NRC Technical Library, Two White Flint North, 11545 Rockville Pike, Rockville, Maryland 20852; telephone: 301-415-7000; email: Library.Resource@nrc.gov. For information on the availability of this material at NARA, visit <https://www.archives.gov/federal-register/cfr/ibr-locations.html> or email fr.inspection@nara.gov. The material may also be obtained from the following sources:

(1) *American National Standards Institute/American Nuclear Society (ANSI/ANS)*, ATTN Standards, 555 N Kensington Avenue, La Grange Park, IL 60526; <https://webstore.ansi.org/>.

(i) ANSI/ANS-6.1.1-2020, “Photon and Neutron Fluence-to-Dose Conversion Coefficients.” IBR approved for § 20.1010(a).

(ii) Reserved

(2) *International Commission on Radiological Protection (ICRP)*, 350 Albert Street Suite 410, Ottawa, Ontario, K1R 1A4, Canada; <https://www.icrp.org/index.asp>.